AME 557 Orbital Mechanics Project 1

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In a distinct effort to avoid the collision of two valuable satellites in low Earth orbit (LEO), our team of engineers was organized to develop an Orbit Determination (OD) algorithm. The algorithm uses satellite observations obtained by a ground-based tracking station at the University of Arizona in Tucson, AZ to compute the satellite's current position and velocity and propagate it forward in time to a specified final epoch. The resulting output from our algorithm provides an accurate prediction of the satellite's future orbital position and velocity and whether an impact with a collaborative team's satellite (Team C) was imminent. To provide a more statistical interpretation of potential impacts, we applied noise to the initial observations to generate a distribution of potential final states at a specified epoch.

I. Nomenclature

μ_E	=	Earth's Gravitational Constant
R_E	=	Radius of the Earth
$ECEF$	=	Earth-Centered-Earth-Fixed Frame of Reference
SEZ	=	Tracking station's local horizon Frame of Reference (Synonymous with ECEF)
ECI	=	Earth-Centered Inertial Frame of Reference
LOS	=	Line of Sight
LST	=	Local Sidereal Time

II. Planetary Constants and Known Values

μ_E = 398600.4354 km^3/s^2
R_E = 6378.1366 km

Site Latitude	32.2227°
Site Longitude	-110.0101°
Site Altitude	757 m

Table 1 Location of Observation Station (University of Arizona)

*M.Sc Aerospace Engineering Candidate, B.Sc in Mechanical Engineering from George Mason University
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Julian Date (JD)	Right Ascension(α)	Declination (δ)	Local Sidereal Time (θ_{LST})
I-Satellite			
2454872.241766892	30.859159090717	79.318796817875	295.996368384251
2454872.241940503	14.564451739639	78.120651560859	296.059039499724
2454872.242114115	0.829762748762	75.903618501209	296.121710615405
C-Satellite			
2454871.514010361	5.931355414284	-26.399712354399	33.286705754698
2454871.514183972	6.369337583606	-23.712111094605	33.349376869963
2454871.514357583	6.814572192903	-20.736087662850	33.412047985644

Table 2 Satellite Observations

III. Introduction

For the purpose of this project, our primary objectives were to utilize the observation data obtained by the Earth-based tracking site to calculate the orbital elements, propagate the position and velocity vectors to a future time, and determine if there was a risk of collision with Team C's satellite within a small region of error. Since we cannot guarantee the accuracy of the collaborative Team's orbit determination and propagation, we decided to also run the algorithm we developed on similar observations of their satellite. By conducting the orbit determination and propagation on both satellites using the same algorithms and code, we mitigated the risk of a multitude of potential calculation variances using other code.

The ground tracking station which provided us with our measurements was located at The University of Arizona with known values mentioned in Table 1. This ground tracking station observed the satellite at three distinct locations in its orbit, providing us with a measure of right ascension, declination, and local sidereal time at an associated Julian date Table 2. Provided this information, we had enough data to utilize an Angles-Only approach to determine the orbit, find the classical orbital elements, and propagate the orbits to a final epoch where collision is most imminent.

Since a multitude of other factors such as solar radiation pressure, atmospheric drag, third-body gravitational influences, and perturbations can have an effect on the true orbit of a satellite, we also simulated a random distribution of 100 errors into the resolved state vectors. These errors in the position and velocity coordinates were then propagated to the final epoch as well. By comparing the approach distances between all these potential orbits of both satellites, we were able to produce a more statistical approach to accounting for errors.

Throughout the project, we had to make decisions on choosing the calculation methods that would provide us with the most accurate orbit modeling and propagation. A vast array of methods and algorithms have been developed, but each has its own advantages and disadvantages. In order to produce the least amount of error in our orbits, we had to determine which method would fit our needs most appropriately. We relied on source information to aid in these decisions, but in many cases, we actually tested multiple approaches through coding and determined which would be most accurate.

IV. Orbit Determination and Propagation

A significant factor in choosing a computational method for determining the orbits was that we were basing our calculations off of line of sight (LOS) observations and did not have range measurements. Thus, we chose to utilize an iterative angles-only method. There are three methods introduced in Vallado[1] for calculating orbits from only angle measurements: Laplacian, Gaussian, and Double R-Iteration. Each method has specific characteristic strengths and weaknesses. As discussed by Vallado, the double r-iteration method is most effective when observations are spread out over a long period of time. Since our observation window was less than one minute, this method was not a suitable candidate for our project. The Gaussian method is most fit to handle observations that have no more than 60° between the first and last observation [1]. It uses data taken at all three measurement points. Hence, this method was chosen as our primary method of angle-only orbit determination.

The third option is the oldest method, developed by Pierre-Simon Laplace in 1780 and has no restriction on the number of observations that can be used. However, the Laplace method was originally developed for celestial objects

well beyond the low Earth Orbits (LEO) where our satellites are located, thus, would likely not be as accurate as the Gauss method for the purposes of our project.

Given the multitude of calculations and visualizations necessary for accurately determining an orbit and its propagation, we chose to use MATLAB® to program our algorithms. This provided a straightforward mechanism for developing a series of algorithms based off the resources cited.

Typically, orbit determination utilizes a slant range measurement distance to the satellite. However, orbit determination using purely optical means of observation require inputs of two angles measured in a topocentric coordinate frame. In this case, the origin of the topocentric coordinate frame was located at the observation station. The observation site provides a measure of right ascension α and declination δ of the satellites with a time stamp of the Julian Date.

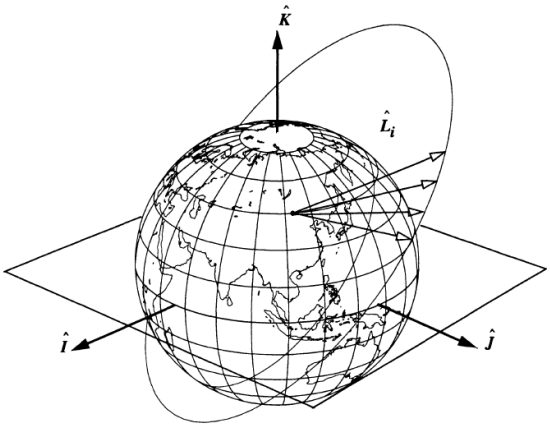


Fig. 1 LOS vectors visualized in the IJK reference frame[1].

Both methods of orbit determination utilize three observations of a satellite made from the ground tracking station. First step for both methods is to form LOS unit vectors at each observation time, $\hat{L}$. Each vector in the matrix corresponds to a specific observation.

$$\hat{L_i} = \begin{bmatrix} cos(\delta_i)cos(\alpha_i) \\ cos(\delta_i)sin(\alpha_i) \\ sin(\delta_i) \end{bmatrix}$$

Given the site location information in Table 1, we are able to calculate a unit vector in the ECI frame of reference that points from the center of mass of the Earth to the observation site.

$$R_{siteECI} = (r_E + site\ altitude) \begin{bmatrix} cos(lat)cos(\theta_{LST}) \\ cos(lat)sin(\theta_{LST}) \\ sin(lat) \end{bmatrix}$$

A. Gauss Method

The Gauss method for orbit determination fits data to all three observation points and is subsequently well-suited for near-Earth satellites. This method can be limited for observations with large angles between them (over 60°) and very well at angles of separation less that 10°. Since the three observations used for this project were closely grouped, this was not a hindrance. This method also assumes three position vectors lie in a single plane (coplanar).

Given that we had no range information to work with, we used the Angles-Only Gauss method, which results in an eighth-degree polynomial, of which you must find the roots in order to determine slant ranges to each satellite. Following Algorithm 46 from Vallado[1]:

The first step was to compute the change in time:

$$\tau_1 = JD_1 - JD_2 \qquad \tau_3 = JD_3 - JD_2$$

Find parameters leading to the middle range magnitude:

$$a_1 = \frac{\tau_1}{\tau_3 - \tau_1} \qquad a_3 = \frac{-\tau_1}{\tau_3 - \tau_1}$$

$$a_{1u} = \frac{\tau_3((\tau_3 - \tau_1)^2 - \tau_3^2)}{6(\tau_3 - \tau_1} \qquad a_{3u} = \frac{\tau_1((\tau_3 - \tau_1)^2 - \tau_1^2)}{6(\tau_3 - \tau_1}$$

Utilizing the inverse of the LOS matrix previously calculated multiplied by our site vectors $[M]$, we can determine the parameters that will be necessary for the eighth-degree polynomial computation:

$$[M] = [L]^{-1}[\vec{r}_{siteECI}]$$

$$d_1 = M_{21}a_1 - M_{22} + M_{23}a_3 \qquad d_2 = M_{21}a_{1u} + M_{23}a_{3u}$$

The eighth-degree polynomial is then found through:

$$C = \hat{L_2} * \vec{r}_{site2}$$

$$r_2^8 - (d_1^2 + 2Cd_1 + r_{site2}^2)r_2^6 - 2\mu(Cd_2 + d_1d_2)r_2^3 - \mu^2d_2^2 = 0$$

After solving for the real root r_2 we can calculate an initial estimate of the slant range values ρ (LOS distance from site to satellite). This process begins with determining the three coefficients c_i:

$$u = \frac{\mu}{r_2^3} \qquad \begin{bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{bmatrix} = \begin{bmatrix} -a_1 - a_{1u}u \\ 1 \\ -a_3 - a_{3u}u \end{bmatrix}$$

We then use these coefficients to gain an initial estimate of ρ from this equation:

$$\begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = \begin{bmatrix} M \end{bmatrix} \begin{bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{bmatrix}$$

Using this initial slant range estimate, we can calculate the position vectors of the satellites.

$$\begin{bmatrix} \vec{r}^1 \\ \vec{r}^2 \\ \vec{r}^3 \end{bmatrix} = \begin{bmatrix} \rho_1\hat{L_1} + \vec{r}_{site1} \\ \rho_2\hat{L_2} + \vec{r}_{site2} \\ \rho_3\hat{L_3} + \vec{r}_{site3} \end{bmatrix}$$

Now that we have obtained the initial estimates, we can use f_i and g_i functions as outlined in Vallado's text (pg 403)[1] to iterate our solution and obtain more accurate slant-range values.

B. Herrick-Gibbs Method

After determining that the angles between vectors was approximately 0.9°, we chose to use the Herrick-Gibbs method acc.[1] to find $\vec{v}_2$ since it is superior to the Gibbs method for angles between vectors less than 1°. This method uses the position vectors and Julian Dates to calculate the velocity vector of the resolved state:

$$\Delta t_{31} = JD_3 - JD_1 \qquad \Delta t_{32} = JD_3 - JD_2 \qquad \Delta t_{21} = JD_2 - JD_1$$

$$\vec{v}_2 = -\Delta t_{32}\left(\frac{1}{\Delta t_{21}\Delta t_{31}} + \frac{\mu}{12r_1^3}\right)\vec{r}_1 + (\Delta t_{32} - \Delta t_{21})\left(\frac{1}{\Delta t_{21}\Delta t_{32}} + \frac{\mu}{12r_2^3}\right)\vec{r}_2 + \Delta t_{21}\left(\frac{1}{\Delta t_{32}\Delta t_{31}} + \frac{\mu}{12r_3^3}\right)\vec{r}_3$$

To build a more flexible software product MATLAB code contains both the Herricks-Gibbs and the Gibbs method. The software calculates the angle between position vectors and uses the appropriate method.

C. Determine Orbital Elements

Now that we had determined $\vec{r}$ and $\vec{v}$ vectors, we now have enough information to calculate the classical orbital elements of the orbit: semi-major axis (a), eccentricity (e), inclination (i), longitude of the ascending node (Ω), argument of periapsis (ω), and true anomaly (f). First step is to calculate the total energy of the orbit (E), using the given value of the gravitational constant for the Earth and the position and velocity vectors calculated. The total energy will always be negative for a bounded orbit.

$$E = \frac{v_0^2}{2} - \frac{\mu_E}{r_0}$$

Knowing the total energy of the orbit, the six classical orbital elements can now be determined[2]:

$$a = \frac{-\mu_E}{2E} \qquad Semimajor\ axis\ [km]$$

$$\vec{e} = \frac{1}{\mu_E}\left[\left(v_0^2 - \frac{\mu_E}{r}\right)\vec{r_0} - (\vec{r_0}\cdot\vec{v_0})\vec{v_0}\right] \qquad Eccentricity\ vector$$

$$e = \|\vec{e}\| \qquad Eccentricity$$

In order to calculate the remaining angles that define the orientation of the orbit and the location of the satellite from perigee, we must calculate the angular momentum. Also, note that IJK are unit vectors of the inertial (ECI) coordinate frame. The angular momentum is defined as the cross product of the $\vec{r}$ and $\vec{v}$. Since these two vectors are coplaner, the angular momentum will be perpendicular to the the orbital plane:

$$\vec{h} = \vec{r_0} \times \vec{v_0}$$

The remaining orbital elements are found using the angular momentum and the IJK unit vectors:

$$i = cos^-1\left(\frac{\vec{K}\cdot\vec{h}}{\|h\|}\right) \qquad Inclination\ [^\circ]$$

$$\vec{n} = \vec{K} \times \vec{h}$$

$$\Omega = sin^-1\left(\frac{\vec{J}\cdot\vec{n}}{n}\right) \qquad Longitude\ of\ Ascending\ Node\ [^\circ]$$

Using Matlab to solve for Ω requires the use of the $atan2$ function (with arguments of $sin(\Omega)$ and $cos(\Omega)$) in order to ensure Ω is in the proper quadrant.

$$\omega = cos^-1\left(\frac{\vec{n}\cdot\vec{e}}{\|\vec{n}\|e}\right) \qquad Argument\ of\ Periapsis\ [^\circ]$$

D. Orbit Propagation

After determining the orbit and orbital elements, we propagated the orbit to a final epoch (Table 3).

Julian Date	2454873.205555555

Table 3 Final epoch to propagate to.

To verify results and allow us a choice we utilized two methods of orbit propagation. The first method uses the built-in MATLAB® function, ODE45 [3]. The second method is a numerical method of solving the Kepler equation. Given the Julian Date of the observation and the final epoch, a flight time can be obtained. With that we can solve Keplers equation for the true anomaly of the final epoch. This allows us to propagate the whole state vector to that time. Both methods were outlined in a lecture by Di Wu [4]. While we decided to implement the former method (ODE45) as our primary option for orbit propagation, we completed both to compare and contrast the accuracy of both.

The result of orbit determination based solely on observations will give initial orbits that must be refined over time with additional observations to achieve greater accuracy. If it is not possible to refine the orbit, it is interesting to note how an error in the initial resolved state will propagate over time. To demonstrate this, we applied a random error of up to $\pm 2\,km$ on the initial position vector and up to $\pm 0.1\,km/s$ on the initial velocity vector. This method generated 100 different states, all of which were propagated to the final Julian date. This gives 100 final states each for the I and C group data. It is interesting to see how many of these states are at a dangerous proximity to one another. In order to determine this, an approach matrix AM was calculated, that contains all 10,000 possible approach distances:

$$AM = \begin{bmatrix} \|\vec{r}_{I,1} - \vec{r}_{C,1}\| & \cdots & \|\vec{r}_{I,100} - \vec{r}_{C,1}\| \\ \vdots & \ddots & \\ \|\vec{r}_{I,100} - \vec{r}_{C,1}\| & & \|\vec{r}_{I,100} - \vec{r}_{C,100}\| \end{bmatrix}$$

This matrix contains all the approach distances for the propagated states. The entries can be sorted and classified according to how close they are.

V. Results

A. Orbit Determination and Propagation

The MATLAB® software we developed for this project requires the planetary constants, site location, and satellite observation data outlined in Section II. In order to complete the Gauss Angles-Only method, the software requires the initial position information of the observation site (latitude, longitude, and altitude), the satellite observations (Julian date, right ascension, declination, and local sidereal time), and the planetary constants (gravitational constant and radius of the Earth). Using this information, we can calculate initial estimates of position vectors to the satellites in the inertial frame, then improve them through iteration as mentioned above in Section IV. Table 4 shows the final state for the I-Satellite

	Initial			**Propagated**		
	I	**J**	**K**	**I**	**J**	**K**
Position (km)	2622	-4782.53	4635.94	-2667.42	3655.95	5538.04
Velocity (km/s)	2.85315	-3.94147	-5.66077	2.80475	-5.08536	4.70083
Iterated Position (km)	2622.02	-4782.42	4636.03	-	-	-
Iterated Velocity (km/s)	2.85333	-3.94177	-5.66119	-	-	-

Table 4 I-Satellite Orbit Determination and ode45 Propagation

Given the data for the I-Satellite, one can repeat the method for the C-Satellite and use both results to calculate a approach distance at the propagated time. This can be done for both implemented methods:

$$ApproachDistance_{ode45} = \|\vec{r}_{f,I,ode45} - \vec{r}_{f,C,ode45}\| = 4320.75\,[km]$$
$$ApproachDistance_{Kepler} = \|\vec{r}_{f,I,Kepler} - \vec{r}_{f,C,Kepler}\| = 2327.25\,[km]$$

The approach distances above are based on position vectors that were both calculated with our code. After data exchange with the C-Group we can calculate the approach distance between our I-Satellite final position and the final position for the C-Satellite as calculated by the C-Group:

$$\vec{r}_{f,C,C-Group} = [-409.1 - 1945.86871.78]^T$$

$$ApproachDistance_{Groups} = \|\vec{r}_{f,I,ode45} - \vec{r}_{f,C,C-Group}\| = 4715.21\,[km]$$

The resolved and iterated position and velocity vector can also be used to calculate the the six classical orbit elements, and thereby establishing the orbit. The classical orbit elements for the I-satellite are shown in table 5

Semi-major Axis (a)	7163.89	km
Eccentricity (e)	.001821	
Inclination	86.4479	deg
RAAN	121.76	deg
Argument of Perigee	75.1551	deg
True Anomaly (f)	64.3864	deg

Table 5 Initial Orbital Elements for I-Satellite

In order to visualize the determined orbit, updated existing code [5] from Embry-Riddle Aeronautical University. The two orbits for the I and C groups can be seen in Fig 2.

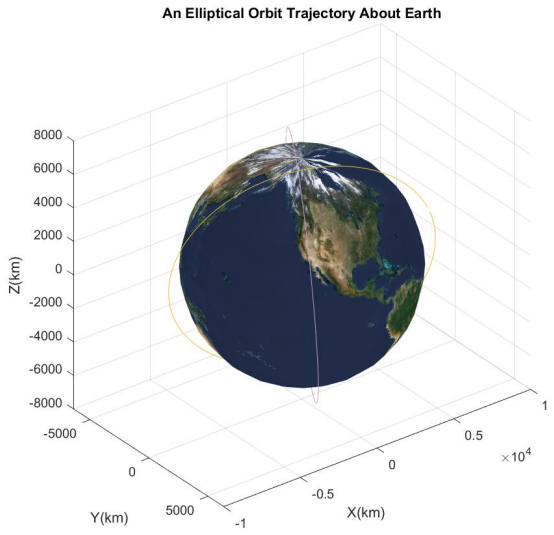


Fig. 2 Plot of the I and C orbits

The orbits cross in the area between northern Siberia and Alaska. This is reasonably close to the collision point specified in section VI.A.

B. Distribution of Propagated Orbits

Using the improved position and velocity vector as a starting point, we added a random error in form of a 3-σ distribution and propagated these new vectors to the final epoch. As mentioned in Section IV we calculated all the obtained approach distances.

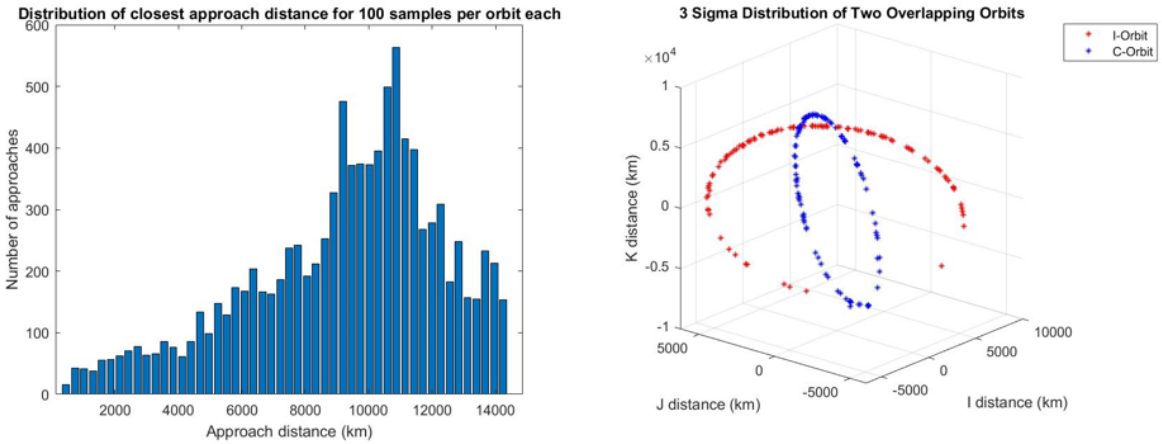


Fig. 3 Distribution of the approach distances.

Fig 3 shows both the number of approach distances between the I and C satellites, and a visual representation of the propagated states. The plots show that a majority of approaches were between the 10,000km and 11,000km with a minimum approach distance of 16km and maximum of 14,000km. For the histogram we divided the range between the minimum and maximum approach distance into 50 sub ranges and summed the number of *AM* entries in that range. It is important to note that these results, will change on every run, unlike the results from Section V.A, due to the random 3-sigma error applied.

The distribution shows however, that this method may not be helpful to determine collision probability. An maximum error of 0.1km/s on the velocity vector components has serious consequences if propagated over an extended period of time. This is the case (e.g. 83,000s propagation time for the I-Orbit) and leads to spread out final states as seen in Fig. 3 after the satellite orbits the Earth approximately 13 times.

C. Comparison of propagation methods

We implemented two propagation methods, Kepler and ODE45, and found that the results vary substantially between them as seen in Table 6. Both methods require the same input from Table 4.

	ode45			**Kepler**		
	I	**J**	**K**	**I**	**J**	**K**
Final Position (km)	-2667.42	3655.95	5538.04	-1862.53	2238.55	6531.21
Final Velocity (km/s)	2.80475	-5.08536	4.70083	3.44097	-5.91322	3.00571

Table 6 Comparison of two propagation methods

We expected the results between the two propagation methods to be much closer and are not completely sure why there is such a variance. One reason might be that ODE45 iterates the dynamic equation whereas the Kepler method iterates to find the future true anomaly and then uses Lagrangian coefficients to produce the propagated position and velocity vectors.
The difference between the two methods is:

$$Distance_{ode45-Kepler} = \|\vec{r}_{f,ode45} - \vec{r}_{f,Kepler}\| = 1835.7\,[km]$$

Fig. 4 shows the distribution plot for both propagation methods. Both plots look very similar, despite the differences shown in Table 6.

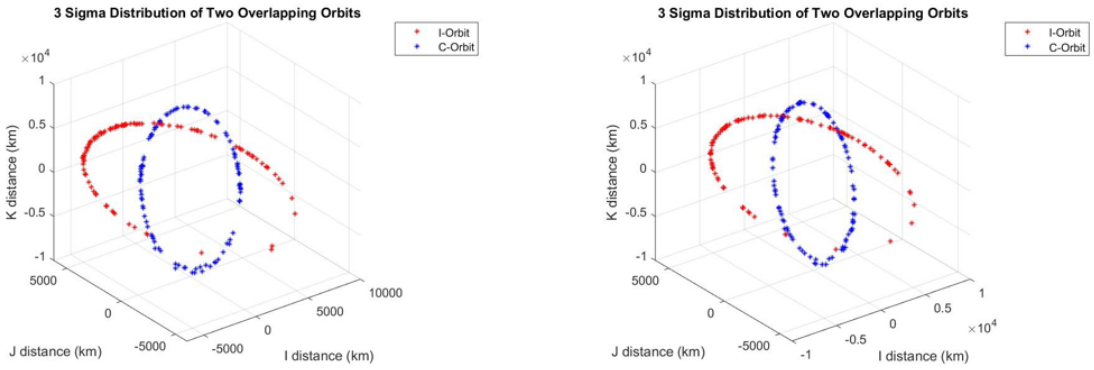


Fig. 4 The Distribution with ode45 propagation (left) and with Kepler propagation (right)

VI. Conclusions

A. Historical Relevance

At 16:56 UTC on February 10, 2009 two communications satellites, Iridium-33 and Kosmos-2251 (translated Cosmos-2251) collided at an altitude of 789 kilometers above Siberia according to the April 2009 Orbital Debris Quarterly Report by the Johnson Space Center [6]. The Kosmos-2251 satellite had been inactive for several years while still maintaining its orbit.

Given strong correlation between the orbit data and collision time and date, it can be deduced that our orbits correspond to these two satellites. This real life application highlights the real significance of orbit determination and propagation and how important it is in maintaining a clean Low-Earth space environment to avoid disastrous scenarios such as the Kessler Syndrome.

B. Summary

The presented report shows how visual observations of a satellite can be used to determine an initial orbit. However, it was shown that the quality of this initial determination needs further refinement to achieve reliable orbit data. This reliability can, in part, be attributed to the chosen method of propagation, as was shown in this report. Additionally, further refinements of the orbit determination and propagation can be made by taking a multitude of factors such as third-body gravitational influences, perturbations, atmospheric drag, solar radiation pressure, etc.

To enhance the accuracy of the orbit determination, additional observations or tracking data should be used to confirm the actual orbit. As mentioned in the section above, the observation data should have led to a collision at the final epoch. However, independent of the method the approach distance was nowhere near a collision. Even the closest approach of the distribution was still enough to not be considered as an imminent collision. The distribution of the initial vectors and their propagation might seem like an useful method to better calculate the risks of an orbit that is not fully determined, but the propagation shows that the error adds up significantly in the time that we should propagate. This leads to data that is not helpful in determining if the collision threat is substantial.

References

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